

Standard method for microCT-based additive manufacturing quality control 5: witness specimen analysis

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Abstract

MicroCT is a well known technique and is often used for non-destructive analysis of parts produced by additive manufacturing. This is especially useful to inspect the parts for potential defects such as excessive porosity or layered defects due to imperfect powder levelling in individual layers. While various online monitoring methods are under development, there is currently not yet any accepted method to identify such defects during the building process. Post-production microCT of the entire part allows for the identification of such errors, but its limitation is the resolution which depends on part size. Hence a large part may have small but important porosities which may be missed by microCT. This can be overcome by using a witness specimen built in the same process, and analysing this smaller sample at a high resolution. The porosity information from the smaller part can be used to evaluate the acceptability of the entire build. The workflow presented here serves to standardize this process: with a simple process, a standard geometry witness specimen and hence standardized scan and analysis procedures. A layered stop-start flaw is shown in the witness specimen to demonstrate the ability to detect this type of flaw.

Introduction

A recent review article highlights the capabilities of present day microCT for the analysis of additively manufactured parts [1]. Although these capabilities are starting to be appreciated more widely in the additive manufacturing community, there is a need for standardization of inspections in order to improve the interpretation and ultimately the proper usage of the technique [2]. To this end we have developed a number of methods for inspection of small coupon samples [3]–[5] and for powder characterization [6]. These can be used for improving processing conditions, prior to building critical parts. In the present method paper, we outline a proposed method of analysing a complex part built by AM and a related standardized witness specimen which must be built in the same build process. In this example the machine was stopped and started 12 hours later, to ensure a stop-start flaw or layered defect is present. These layered defects have been detected by microCT scans as reported in [7].

The method

The part itself is initially scanned in a typical microCT system [8] using optimization procedures as outlined in [9] and as shown for AM parts in [10]. The whole part, its witness specimen and its 10 mm coupon cube is shown in Figure 1. The witness specimen is a 15 mm diameter cylinder built vertically up to total height of the complex part height. In this case the complex part was a topology optimized bracket designed for a load-bearing application.



Figure 1: Photograph of the complex part – a topology optimized bracket, its witness specimen and 10 mm coupon cube

The testing methodology is as follows: (a) first make a high quality microCT scan of the entire bracket, (b) then make a (higher) resolution scan of the witness specimen in order to visualize thin layered defects if present, and (c) an optional high resolution “zoom” scan of the expected location of defects in the complex part. In this case layered defects were introduced on purpose, to demonstrate the capability to detect them in the witness specimen. This entire workflow is demonstrated in this paper.

For the high quality scan of the part itself, the resolution is limited by geometric magnification to the sample width, in this case the resolution was $46\text{ }\mu\text{m}$. The sample was loaded at approximately a 45 degree angle, and scanned using 200 kV, 100 μA with a 0.5mm copper filter. Image acquisitions were 250 ms each, with 3000 images recorded during a 360 degree rotation. This scan was done without averaging to allow higher throughput of samples, while more averaging or longer image acquisitions can lead to higher quality image data (meant to detect smaller pores, for example). In this case the scan time was approximately 20 minutes. Image analysis was done in VGStudioMax 3.1. An automatic surface determination function was applied, followed by a default porosity analysis which

found no detected pores, as confirmed by manual inspection of slice images. Manual inspection revealed some surface imperfections such as that shown in Figure 2.

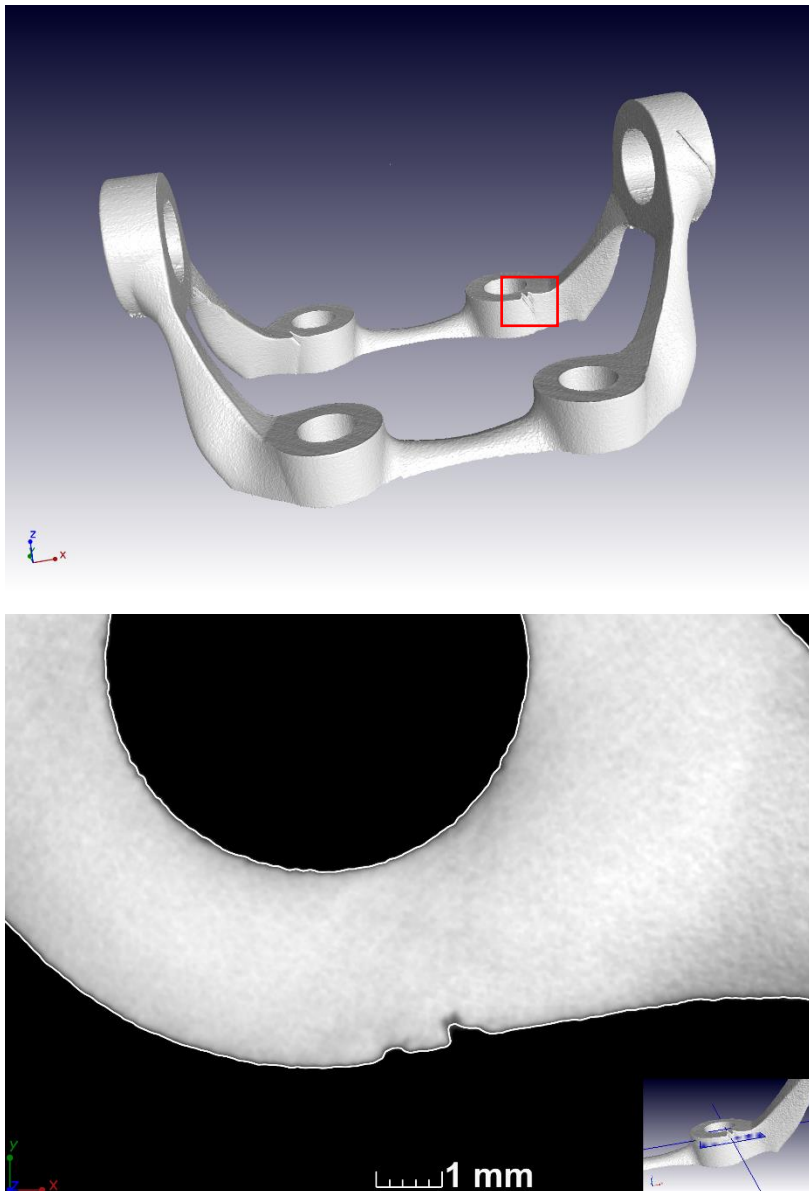


Figure 2: MicroCT scan of whole part – no internal voids were found, but surface manufacturing errors could be identified such as the one shown in red square in (a) and a close up slice view of this area in (b).

In this case, despite a stop-start cycle of 12 hrs, no layered defects were found inside the complex part at the 46 μm resolution. A 20 μm scan of the witness specimen revealed the presence of layered defects as shown in Figure 3. This scan was done with the witness specimen loaded at a 45 degree angle with identical settings as the previous scan, besides resolution improvement. This indicates the ease of detecting layered defects in a witness specimen in a short scan time. For analysis, a simple slice image reveals the layered porosity.



Figure 3: Witness specimen scan at 20 μm , showing layered flaw due to stop-start during the build: (a) 3D visualization of layered porosity, (b) in-plane slice view, (c) cross-sectional (side) slice view.

In order to demonstrate an even higher resolution witness specimen scan, the sample was loaded at less of an angle and scanned with similar settings but including averaging to improve image quality. In this case at a higher magnification and resolution of 10 μm , the part was scanned in 4 parts and stitched together automatically during reconstruction. This total scan time was 2 hours and the improved resolution shows the layer defect more clearly as seen in Figure 4. This longer scan is

probably preferred due to the ability to detect potentially smaller defects, but only one of the two types of scans shown here are required.

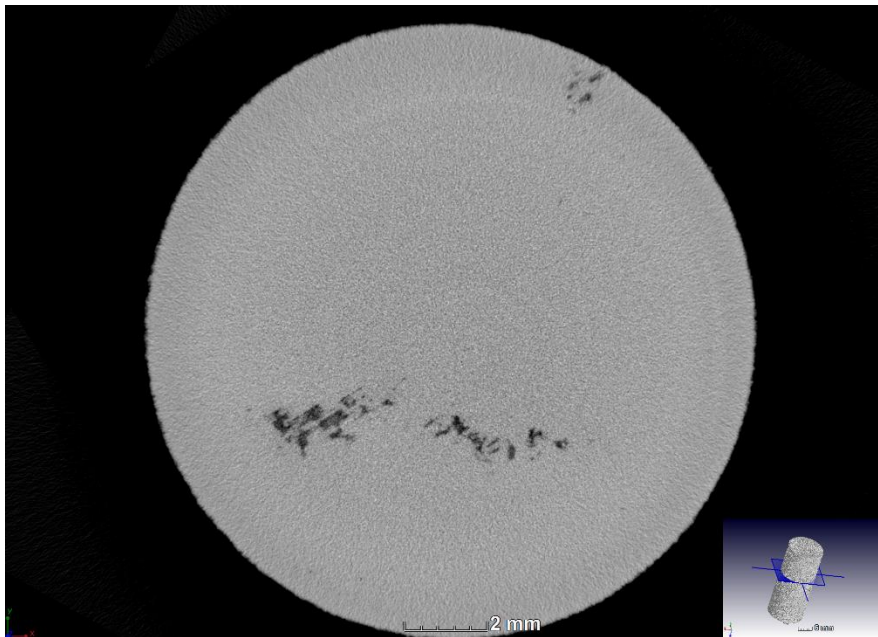


Figure 4: High resolution scan of witness specimen showing more clearly the layer defects (10 μm resolution)

A higher resolution scan at 23 μm of the potentially problematic areas in the complex part is shown in Figure 5. The potentially problematic areas in the complex part were identified based on the build height and the complex part build orientation which was known in this case. However, no layer defects were found in this case. Layered stop-start flaws are not found everywhere in the build plane, and in this case they were found in the witness specimen but not the part itself. The idea of this process is to use witness specimens in order to find such unwanted defects, especially when the complex part does not allow high resolution scans of arbitrary sections. Further work will be required to ensure these flaws are not missed in witness specimens and at what resolution exactly they will be visible in complex parts. Here we focus on the methodology and standard process for witness specimen scans allowing up to 10 μm voxel size.

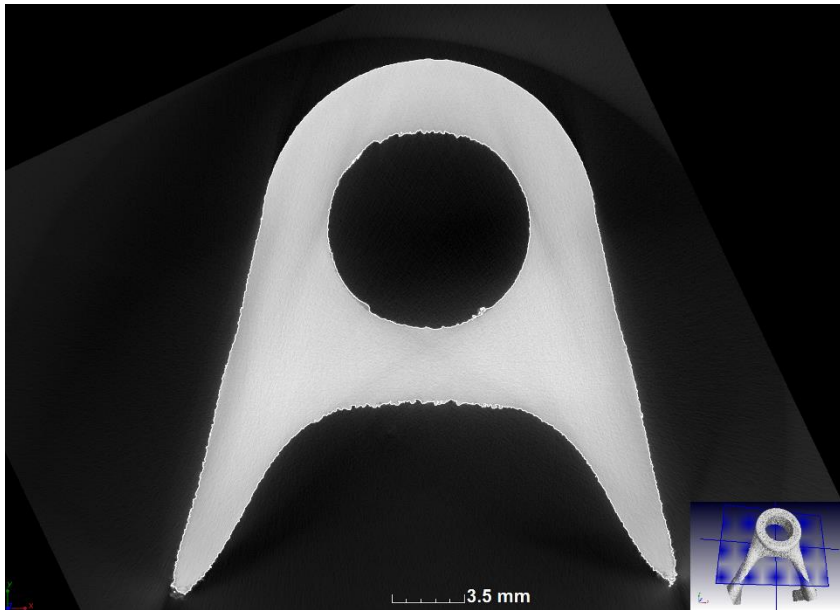


Figure 5: High resolution “zoom scan” of potentially problematic area, corresponding to layer defect height in build. No layer defects found in this case.

Conclusion

We have demonstrated a method and workflow for microCT scanning and image analysis to detect the presence of stop-start flaws (layered void regions) in additive manufactured witness specimens. This method should be useful for routine analysis with the only modification being the resolution of the scan of the complex part, which depends on part size. The standardization of witness specimen diameter allows a fixed methodology for all additive manufactured witness specimens according to the guidelines presented here. Larger parts will have longer witness specimens and this will require longer scan times, this is the only limitation. The advantage is that potentially only one witness specimen is required for an entire batch of parts built, to ensure lack of layered or other unwanted types of defects.

Supplementary material

A video is included which demonstrates the image analysis workflow step by step. The analysed volume data is available and will be published online and made freely accessible.

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